Development of a Clinical Question-Answering Corpus with Realistic Multi-Answer Challenges

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Abstract: We generated a clinical question-answering (QA) corpus that involves realistic, multi-answer cases by converting the concept-relation annotations from the i2b2 Adverse Drug Event extraction challenge dataset. A total of 91,440 QA entries were derived, representing a diverse array of 1-to-0, 1-to-1, N-to-1, and 1-to-N relations between the question and answer(s).

Introduction: Clinician's thought processes and decision-making often involve a cascade of questions and answers. Achieving human-like question-answering (QA) capability is highly regarded in artificial intelligence (AI). Medical QA research has therefore gained vital momentum over the past decade. A new generation of AI scientists is updating the "state-of-the-art" at a daunting pace almost every few months, if not weeks. One of the sought-after applications is to find the answer within a given document (a.k.a. reading comprehension), which enables patient-specific QA based on the information mentioned in the clinical text. As an essential component in AI engineering, QA training data determines the likelihood of success not only in terms of annotation quality but the fidelity of representing the target scenario. Along with other issues observed in existing medical QA corpora,[1] the mainstream annotation approach knowingly simplifies the task into a one-answer-one-document scheme. The simplification has practically made development and evaluation easier for promoting the initial growth of the field. However, it is unrealistic in the sense that medical QA can naturally have multiple qualified answers within one document – sometimes all need to be captured to sufficiently answer a question. To address this gap, we created a clinical QA corpus that involves realistic multi-answer cases by converting the concept-relation annotations from an existing AI challenge dataset.

Methods: The 2018 i2b2 Adverse Drug Event (ADE) challenge[2] was originally organized for extracting ADEs and various drug-related entities in clinical text. The dataset contains a gold standard annotation of 83,869 concepts and 59,810 relations in 505 discharge summaries. Given our primary interest in why-QA, we focus on generating QA pairs from the subset of annotated Reason-Drug relations. Each Reason-Drug relation has two arguments: one Reason concept and one Drug concept. Accordingly, we can generate a question from a Drug, e.g., "Why morphine was prescribed to the patient?", and its Reason "pain" can then be made the answer. Since there exist many-to-many relations across the Reason and Drug concepts in the dataset, we can derive the corresponding N-to-1 or 1-to-N QA entries as a logical result. In addition, by leveraging the fact that some Drug concepts do not have any Reason in the text, these no-reason instances allow us to generate unanswerable questions – a critical subset in QA challenges for evaluating AI system specificity. As another value-adding feature, we inject paraphrastic questions for enhancing the generalizability of the potentially trained systems.[3]

Results & Discussion: The table below summarizes the QA categories and examples in our generated corpus. We obtained a total of 91,440 QA entries, of which half (51%) are unanswerable. The answerable ones include a substantial percentage of 1-to-1 (24%), N-to-1 (15%) and 1-to-N (10%) QAs, and each has more than 9,000 entries. The results are formatted into the widely adopted Stanford Question Answering Dataset (SQuAD) 2.0 JSON, except that we now allow each 1-to-N QA to have a list of reasons under the answer block. As the major contribution of this work, these diverse and sizable annotations represent more realistic train/test data for developing robust clinical QA systems that will transcend the conventional, limited single-answer task definition. We are in the process of engaging our domain experts to quality-audit the corpus and will share it with the research community.

Category	Frequency (%)	Example	Processing method
No reason annotated (1-to-0 QA)	46,278 (51%)	**Mirtazapine** 15 mg PO QHS [*only the drug is mentioned but no reason is documented*]	Make it an unanswerable QA entry
1 drug 1 reason (1-to-1 QA)	21,906 (24%)	The patient received **morphine** for *pain* as needed	Make into a 1-to-1 QA entry
N drugs 1 reason (N-to-1 QA)	14,004 (15%)	*Hypertension*: Severely elevated blood pressure. Started **amlodipine**, **metoprolol**, and **isorbide**.	Break into N separate 1-to-1 QA entries
1 drug N reasons (1-to-N QA)	9,252 (10%)	**albuterol sulfate** 90 mcg… Puff Inhalation Q4H for *sob* or *wheeze*	List the N reasons under answer block to form a 1-to-N QA entry

References

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